SET-THEORETICALLY INVARIANT MORPHISMS INTO LOCALLY SEPARATED TARGETS ARE INVARIANT WHEN THE RELATION IS REDUCED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids-quotients.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The reducedness of R is essential; without it the bijective closed immersion Z to R need not be an isomorphism.

Editorial note. The label lemma-set-theoretically-invariant-invariant-when-reduced names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter Definition definition-set-theoretically-invariant (navigation stub): set-theoretically R-invariant means U(k) to X(k) equalizes s and t for every algebraically closed k over B; equivalently the equalizer Z below maps surjectively onto R on topological spaces, since every geometric point of R lifts to a k-point
- X locally separated over B: the diagonal of X over B is an immersion, so the equalizer Z of the two maps from R to X times over B X, taken over the diagonal, is an immersion into R
- Schemes, Lemma schemes-lemma-immersion-when-closed: an immersion whose image is closed is a closed immersion; in particular a bijective immersion is a closed immersion
- A bijective closed immersion into a reduced algebraic space or scheme is an isomorphism: its defining quasi-coherent ideal sheaf consists of nilpotent sections and hence vanishes on a reduced target

Source reference. Original reference 048V: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 953–953. Prerequisite pointer invoked at 1002; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01IQ: sources/stacks-chow/tags.txt, source lines 1987–1987. Prerequisite pointer invoked at 1021; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation of Definition 048V. Let $\phi : U \to X$ be a morphism of algebraic spaces over B. Assume*

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(1) *ϕ is set-theoretically R-invariant,*
(2) *R is reduced, and*
(3) *X is locally separated over B.*

Then ϕ is R-invariant.

Proof. Consider the equalizer

$$Z = R \times_{(\phi,\phi)\circ j, X\times_B X, \Delta_{X/B}} X$$

algebraic space. Then $Z \to R$ is an immersion by assumption (3). By assumption (1) $|Z| \to |R|$ is surjective. This implies that $Z \to R$ is a bijective closed immersion (use Schemes, Lemma 01IQ) and by assumption (2) we conclude that $Z = R$. $\square$